Topic: (draft) App Designer File Formats

References

- To replace: Compatibility Between Different Releases of App Designer - MATLAB & Simulink
- Live Code File Format (.mlx) - MATLAB & Simulink
- Live Code File Format (.m) - MATLAB & Simulink
- Plain Text Beta Documentation

- [x] check rules for guibutton/label/menuitem

App Designer File Formats

There are two file formats for storing apps in App Designer. By default, App Designer stores apps using a .mlapp extension. This type of app can only be edited in App Designer.

You can also save apps using a plain text file format that consists of two files (.m and .xml). The plain text format is useful for apps that are managed with source control, or that you would like to edit outside of App Designer (since R2026b).

Choose App File Format

For the most part, apps saved in the plain text format and the binary format have the same behavior. This table summarizes the main considerations when choosing a file format for your app.

Use Case	File Format
Share app with users across multiple MATLAB releases	binary (.mlapp)
Use source control	plain text (.m + .xml)
Simulink apps	binary (.mlapp)
User authored components	binary (.mlapp)

	Plain Text Apps (since R2026b)	Binary Apps
File Extensions	• .m and .xml	.mlapp
Editors	• App Designer • MATLAB Editor • External editors	• App Designer
Source Control	• MATLAB Comparison Tool • Source Control Integration in MATLAB - MATLAB & Simulink	• MATLAB Comparison Tool (link to: Compare and Merge Apps - MATLAB & Simulink)
Limitations	Not compatible with: • Simulink apps • User authored components	Not compatible with: • MATLAB Editor • External editors • External control tools

Modify Default App File Format

By default, new apps created in App Designer are stored with an .mlapp extension. You can change the default app file format.

Convert App File Format

You can convert existing apps into a different file type by creating a copy of your app.

Binary Apps

The default app file format consists of one file with a `.mlapp` extension. This app format uses Open Packaging Conventions technology, which is an extension of the zip file format.

Apps stored with a `.mlapp` extension can only be edited in App Designer.

Plain Text Apps

The plain text app file format consists of two files:

- A MATLAB® code file (`.m`) that contains all the app code you write, such as callbacks and properties.

- An XML file (`.xml`) that contains the app configuration, which includes the app components, layout, and metadata.

App code and app configuration files can be opened in both App Designer and external editors.

App Code File (.m)

The MATLAB code file (`.m`) associated with your app stores the app code. This code includes:

- Properties, including both App Designer-generated properties for each UI component in your app and any properties you create to share data in your app.
- Callback functions
- Helper functions

The file also contains a property, `AppConfigFilename`, that stores the path to the XML file with your app layout.

The recommended way to edit this file is in App Designer **Code View**. If you open the code file from the MATLAB Files pane, it will automatically open into App Designer **Code View**.

Alternatively, you can open the code file in other editors. For example, to open the file in the MATLAB Editor, right-click the file in the Files pane and select **Open as Text**. However, if you edit this file outside of App Designer, you might lose code when you load the app in App Designer in the future.

You can use other editors for these tasks:

- Viewing the app code outside of App Designer
- Adding code to sections of the file that are editable in App Designer **Code View**. For example, you can make changes to the code in your callback functions or in property blocks you manage.

Using other editors for these tasks may result in losing code when you load the app in App Designer in the future:

- Making changes to the app layout
- Adding a new callback function
- Adding a new property to the UI components block

App Configuration File (.xml)

The XML file associated with your app stores the app configuration. This configuration includes:

- App layout and UI components

- Component configuration, such as any properties you set on a UI component in your app
- App details, such as the app name and author
- Internal App Designer data used to load and run your app

App Designer is the recommended environment for editing apps. When you interactively develop your app in App Designer, for example by adding a new component or updating the properties of a component, App Designer updates the XML file to reflect the changes.

Compatibility Between Different Releases of App Designer

If you develop apps over multiple releases of MATLAB, or if you edit or run a shared App Designer app in a different release than it was saved in, App Designer makes updates to your app to provide compatibility between the releases. If you encounter errors, you can open the app in App Designer to help resolve them.

You can now save apps using a plain text app format that consists of two files (.m and .xml). To run these apps in earlier versions of MATLAB, convert to the binary app format (.mlapp).

Edit and Run Apps in Later Releases

(same)

Edit and Run Apps in Earlier Releases

If you have an existing `.mlapp` app that was created in one MATLAB release and you are using an earlier MATLAB release, you can open the app in App Designer for editing.

App Designer removes any functionality that is unsupported in the release you are using and provides an option to see the differences in the updated app. This removal might impact the app functionality. For example, if you load an app with a UI component in a release before the component was introduced, App Designer removes the component from the app.

If you try to run an app in an earlier release, you might encounter errors if the app uses functionality that is not supported in your release. To resolve the errors, open the app in App Designer and view the changes App Designer makes to the app.

Existing apps stored in the plain text file format will not open in App Designer in MATLAB releases before R2026b. If you try to open the app code file or app configuration file, then it will

open in the MATLAB Editor. You will not be able to run your app unless you switch to a supported release.

Save Copy of App

App Designer provides two options to create copies of your app: **Save Copy As** and **Save As**. You can access both options from the **Save** button in the App Designer toolstrip. These options serve different purposes, and their behavior is also different.

- To save a copy of your app to edit or share later, use **Save Copy As**. When you use this option, App Designer saves the copy of the app in the specified folder, but it does not replace the app in your current session.
- To save a copy of your app and continue editing it, use **Save As**. When you use this option, App Designer saves the copy of the app in the specified folder and replaces the app in your current session.

You can convert an existing MLAPP app to the plain text format, and vice versa, by creating a copy and selecting either `MATLAB APP (*.m)` or `MATLAB APP (*.mlapp)` when prompted.

First Draft

App Designer File Formats

notes

- **Overview of M+XML and MLAPP files**
  - **Breakdown of how user code vs. generated code is represented**
  - **Decision-making support about when to use each, including what isn't supported (e.g. UAC? Simulink apps?)**
  - need to draft subtitle?
    - live script pages don't have subtitles; they are listed as bullet points under "what is a live script?" page

By default, App Designer stores apps using a binary file format with a .mlapp extension. The binary file format consists of one file and can only be edited in App Designer. This binary app format uses Open Packing Conventions technology, which is an extension of the zip file format.

You can also save apps using a plain text app file format (.m and .xml). The plain text format is useful for apps that you use with source control or that have code that you would like to edit outside of App Designer (*Since R2026b*).

The plain text app format is not compatible with Simulink apps or user authored components.

Both app file formats have the same behavior in App Designer. For information on creating app layouts interactively, see Lay out Apps in App Designer Design View. For information on editing app code, see Manage Code in App Designer Code View.

MLAPP

notes

- **MLAPP: How code is represented**
  - **All code in MLAPP file**
  - **User code writeable, layout and data read-only**
  - Worth mentioning OPC(?), like .mlx file page does? Live Code File Format (.mlx) - MATLAB & Simulink
  - maybe nix this section? not much to say unless it's in comparison to the plain text format (table? list?)

The binary file format consists of one file (.mlapp) and can only be edited in App Designer. This binary app format uses Open Packaging Conventions technology, which is an extension of the zip file format.

M+XML

notes

- **M+XML: How code is represented**

The plain text file format consists of two files:

- A MATLAB code file (.m) that contains all of the app code you write, such as callbacks and properties.
- An XML file (.xml) that contains the app configuration, which includes the app components, layout, and metadata.

MATLAB Code File (.m)

notes

- **User code in M file**
- Note needed about the warning that pops up?

The MATLAB code file (.m) associated with your app stores the app code. This code includes:

- Properties, including both App Designer-generated properties for each UI component in your app and any properties you create to share data in your app.
- Callback functions
- Helper functions

The file also contains a property, AppConfigFilename, that stores the path to the XML file with your app layout.

The recommended way to edit this file is in App Designer **Code View**. If you select the code file from the MATLAB file tree, it will automatically open into App Designer Code View.

Alternatively, you can open the file in other editors. For example, to open the file in the MATLAB Editor, right-click the file in the Files pane and select **Open as Text**. However, if you edit this file outside App Designer, you might lose code when you load the app in App Designer in the future.

You can use other editors for these tasks:

- Viewing the app code outside of App Designer
- Adding code to sections of the file that are editable in App Designer **Code View**. For example, you can make changes to the code in your callback functions or in property blocks that you manage.

Using other editors for these tasks may result in losing code when you load the app in App Designer in the future:

- Making changes to the app layout
- Adding a new callback function
- Adding a new property to the UI components properties block

XML File

notes

- **Layout and data in XML file**
  - **Component and properties as elements**
  - **Design-time data as attributes**
  - **Metadata**

The XML file associated with your app stores the app configuration. This configuration includes:

- App layout and UI components
- Component configuration, such as any properties you set on a UI component in your app
- App details, such as the app name and author
- Internal App Designer data used to load and run your app

App Designer is the recommended environment for editing apps. When you interactively develop your app in App Designer, for example by adding a new component or updating the properties of a component, App Designer updates the XML file to reflect the changes.

Modify Default App File Format

notes

- **How to modify your default file format**
- is there a way to do this from the command line?
- rewrite to be more general

In MATLAB, on the **Home** tab, in the **Environment** section, click **Settings**. Select App Designer in the left pane. Under **Create new apps as:**, select **M**. Then, select **Blank App** from the App Designer Start Page to create a new app with the plain text file format.

By default, new apps created in App Designer will be stored with an .mlapp extension. You can change the default app file format.

1. In MATLAB, on the **Home** tab, in the **Environment** section, click **Settings.**
2. Select App Designer in the left panel.
3. Under **Create new apps as:**, select either **M** or **MLAPP**.

Convert App File Type

notes

- **How to convert apps between formats**
  - **Preferences** → what does this mean
  - **Save As**
  - A note about shadowing?
  - combine this with Save Copy of App → add note about changing file type

You can convert an existing MLAPP file to the plain text format, and vice versa.

1. Open the app in App Designer.
2. In the **Designer** tab, select **Save > Save As** or **Save a Copy As**.

3. Rename the app, and then under **Save as type:**, select either **MATLAB APP (*.m)** or **MATLAB APP (*.mlapp)**.
4. Click save to create a copy of your app with the chosen format.

Compatibility Between Different Releases of App Designer

notes

- **Compatibility of different App Designer files**
  - **Running/editing/sharing MLAPP files between releases** → same info as before
  - **Running/editing/sharing M+XML files between releases** → earlier versions recognize app code (.m) as a class, can be opened in MATLAB but will not run